Exp 2: Iris Dataset Preprocessing

AIM:

To preprocess the Iris dataset by encoding categorical variables, scaling numerical features, and splitting the dataset into training and testing sets for machine learning applications.

PROCEDURE:

1. Import the required Python libraries such as Pandas, Seaborn, and Scikit-learn.
2. Load the Iris dataset using the Seaborn library.
3. Display the dataset and analyze its structure, statistical summary, and missing values.
4. Encode the **Species** column into numerical values using Label Encoding.
5. Separate the dataset into input features and target labels.
6. Standardize all numerical features using StandardScaler.
7. Split the dataset into training and testing sets using the train-test split method.
8. Display the shapes of the training and testing datasets.
9. Verify that the data is properly preprocessed and ready for machine learning algorithms.

CODE:

```
import pandas as pd

import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder, StandardScaler
```

```
df = sns.load_dataset("iris")

print(df.head())
print(df.info())
print(df.describe())
print(df.isnull().sum())

encoder = LabelEncoder()

df["species"] = encoder.fit_transform(df["species"])

X = df.drop("species", axis=1)
y = df["species"]

scaler = StandardScaler()

X = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42
)
```

```python
print("Training Data Shape:", X_train.shape)

print("Testing Data Shape:", X_test.shape)

print(X_train[:5])
```

OUTPUT:

```
...    sepal_length  sepal_width  petal_length  petal_width species
    0           5.1          3.5           1.4          0.2  setosa
    1           4.9          3.0           1.4          0.2  setosa
    2           4.7          3.2           1.3          0.2  setosa
    3           4.6          3.1           1.5          0.2  setosa
    4           5.0          3.6           1.4          0.2  setosa
    <class 'pandas.core.frame.DataFrame'>
    RangeIndex: 150 entries, 0 to 149
    Data columns (total 5 columns):
     #   Column        Non-Null Count  Dtype  
    ---  ------        --------------  -----  
     0   sepal_length  150 non-null    float64
     1   sepal_width   150 non-null    float64
     2   petal_length  150 non-null    float64
     3   petal_width   150 non-null    float64
     4   species       150 non-null    object 
    dtypes: float64(4), object(1)
    memory usage: 6.0+ KB
    None
           sepal_length  sepal_width  petal_length  petal_width
    count    150.000000   150.000000    150.000000   150.000000
    mean       5.843333     3.057333      3.758000     1.199333
    std        0.828066     0.435866      1.765298     0.762238
    min        4.300000     2.000000      1.000000     0.100000
    25%        5.100000     2.800000      1.600000     0.300000
    50%        5.800000     3.000000      4.350000     1.300000
    75%        6.400000     3.300000      5.100000     1.800000
    max        7.900000     4.400000      6.900000     2.500000
    sepal_length    0
    sepal_width     0
    petal_length    0
    petal_width     0
    species         0
    dtype: int64
    Training Data Shape: (112, 4)
    Testing Data Shape: (38, 4)
    [[-1.02184904  1.24920112 -1.34022653 -1.3154443 ]
     [-0.7795133   2.40018495 -1.2833891  -1.44707648]
     [-0.05250608 -0.82256978  0.76275827  0.92230284]
     [ 0.18982966  0.78880759  0.42173371  0.52740629]
     [ 1.03800476  0.09821729  0.53540856  0.3957741 ]]
```

RESULT:

The Iris dataset was successfully preprocessed by encoding the target variable, scaling the numerical features, and splitting the data into training and testing sets. The processed dataset is ready for building and evaluating machine learning models.